

- 1.
 2. A. Data Valuation
 3. B. LLM
 4. C. Active Learning
 5. D.
 6. E. SCX
 - 7.
-

1.

Web + + 30

SCX SCX

SCX

- $SCX_m(s) = P((f_m(x), y) < \cdot \mid x = s) \text{ ---}$
 - $V(s) = r(s) - L(s) [1 - D(s)] \max_m SCX_m(s) \text{ ---}$
 - $s: X \rightarrow S/$
 - state-wise acquisition point-wise acquisition
-

A. Data Valuation

A.1 Data Shapley

A.1.1 Data Shapley

- Ghorbani & Zou, “Data Shapley: Equitable Valuation of Data for Machine Learning”, ICML 2019
- <https://arxiv.org/abs/1904.02868>

””

Shapley i

$$\phi_i = (1/N!) \sum_{S \ni i} [v(S \cup \{i\}) - v(S \setminus \{i\})]$$

$$v(S) = \sum_{i \in S} \phi_i$$

SCX 1. **vs** Data Shapley SCX — s 2. **vs** Data Shapley $v(S)$ SCX 3. Data Shapley $O(2^N)$ MC $O(N^2)$

SCX - Data Shapley ”””” — Shapley SCX $C(s)$ - Data Shapley — SCX - Data Shapley SCX

A.1.2 Monte Carlo Shapley (TMC-Shapley)

- Ghorbani & Zou, ICML 2019
- Shapley MC

$$O(N^2) \quad O(N \log N)$$

SCX TMC SCX state-wise valuation

A.1.3 KNN-Shapley

- Jia et al., “Towards Efficient Data Valuation Based on the Shapley Value”, AISTATS 2019
- <https://arxiv.org/abs/1902.10275>

KNN surrogate model KNN $O(n \log n)$ Shapley

SCX - KNN-Shapley KNN SCX - KNN curse of dimensionality SCX +

A.1.4 CHG Shapley

- Li et al., “CHG Shapley: Efficient Data Valuation and Selection towards Trustworthy Machine Learning”, 2024

- <https://arxiv.org/abs/2406.11730>

Compound Hardness & Gradient (CHG) epoch Shapley
marginal contribution

SCX CHG SCX state-conditioning — A B

A.1.5 Fast-DataShapley

- “Fast-DataShapley: Neural Modeling for Training Data Valuation”,
ACM WSDM 2026
- <https://arxiv.org/abs/2506.05281>

Shapley

2

SCX Fast-DataShapley Shapley — SCX /

A.1.6 Asymmetric Data Shapley (ADS)

- “Rethinking Data Value: Asymmetric Data Shapley for Structure-Aware
Valuation”, 2025
- <https://arxiv.org/abs/2511.12863>

Shapley /

SCX ADS “asymmetric” A B A BSCX “state-conditioned” —

A.1.7 Geometric Data Valuation via Leverage Scores

- “Geometric Data Valuation via Leverage Scores”, 2025
- <https://arxiv.org/abs/2511.02100>

Shapley leverage score dummyefficiency symmetry

SCX Leverage scores ”” — SCX (s) SCX

SCX

A.2 Influence Function

A.2.1 Influence Functions

- Cook & Weisberg, “Residuals and Influence in Regression”, 1982
- ML Koh & Liang, “Understanding Black-box Predictions via Influence Functions”, ICML 2017
- <https://arxiv.org/abs/1703.04730>

Hessian

$$I(\mathbf{x}_i, \mathbf{x}_{\text{test}}) = - \mathbf{L}(\mathbf{x}_{\text{test}})^T \mathbf{H}^{-1} \mathbf{L}(\mathbf{x}_i)$$

””

SCX 1. **vs** Influence Functions SCX 2. **vs** IF -SCX 3. **Hessian** IF
Hessian LiSSA, CG

SCX - IF - IF $\mathbf{x}_{\text{test}}$ SCX - IF

A.2.2 TRAK (Tracing with the Randomly-Projected After Kernel)

- Park et al., “TRAK: Attributing Model Behavior at Scale”, ICML 2023
- <https://arxiv.org/abs/2212.02522>
- <https://github.com/JD-ETH/trak>

NTKNeural Tangent Kernel

$$(\mathbf{x}_i, \mathbf{x}) = \langle \mathbf{x}_i, \mathbf{M}^{-1}(\mathbf{x}) \rangle$$

$$(\mathbf{x}_i) = \mathbf{J}(\mathbf{x}_i) \mathbf{P} \mathbf{M} \mathbf{NTK}$$

2-3

SCX TRAK TRAK SCX —

A.2.3 D-TRAK

- Zheng et al., “Intriguing Properties of Data Attribution on Diffusion Models”, ICLR 2024
- <https://github.com/qwkkkkk/D-TRAK>

TRAK

SCX

A.2.4 LoGra / LogIX

- Choe et al., “What is Your Data Worth to GPT? LLM-Scale Data Valuation with Influence Functions”, NeurIPS 2025
- <https://arxiv.org/abs/2405.13954>

Kronecker LoRA $O(nk)$ $O((nk))$ LoGra + LogIX

Llama3-8B 1B token 6500x 5x GPU

SCX LoGra LLM SCX state-conditioned

A.2.5 Generalized Group Data Attribution (GGDA)

- “Generalized Group Data Attribution”, 2024
- <https://arxiv.org/abs/2410.09940>

TRAK/IF/TracIn 10x-50x

SCX GGDA ”” SCX ””/GGDA SCX

A.3 LOO /

A.3.1 Leave-One-Out

$LOO_i = v(D) - v(D_{-i})$

vs i

- $O(N)$ - LOO -

A.3.2 Kairos

- “Kairos: Scalable Model-Agnostic Data Valuation”, 2025
- <https://arxiv.org/abs/2506.23799>

MMD LOO $O(mN)$ $O(1/N^2)$ Data Shapley 50

SCX Kairos

A.4 pyDVL

- <https://github.com/aai-institute/pyDVL>
- <https://pydvl.org>
- v0.10.0 (2024)

- Shapley MC class-wise - Banzhaf MSR Banzhaf - Beta Shapley - Least Core MC - LOOData-OOBKNN ShapleyOwen - DirectInfluenceCGBlock CGLiSSAArnoldiEKFACNystromDataInf

SCX pyDVL — **SCX** **pyDVL** pyDVL state-wise acquisition

B. LLM

B.1 DataComp-LM (DCLM)

- “DataComp-LM: In search of the next generation of training sets for language models”, NeurIPS 2024
- <https://arxiv.org/abs/2406.11794>
- <https://datacomp.ai/dclm/>

benchmark—

model-based filteringDCLM-Baseline (7B) MMLU 64% 2.6T token Mistral-7B63% Llama 3 8B66% 6.6

SCX DCLM benchmark — **SCX** **SCX** ”” DCLM trackDCLM 53 **SCX**

SCX - DCLM web **SCX** / - DCLM perplexity heuristic **SCX** - **SCX** DCLM
””””

B.2 LESS (Low-rank gradiEnt Similarity Search)

- Xia et al., “LESS: Selecting Influential Data for Targeted Instruction Tuning”, ICML 2024
- <https://arxiv.org/abs/2402.04333>
- <https://github.com/princeton-nlp/LESS>

1. LoRA adapter → Adam LoRA → 2. → / 3. Adam

- 5% LESS - LLaMA-2-7B LLaMA-2-13B, MISTRAL-7B - 2-5 MMLU, TyDiQA, BBH

arXiv:2410.16208 - LESS perplexity - BM25 embedding 7B-13B - 70B LESS

SCX 1. vs LESS SCX / 2. vs LESS SCX / 3. LESS SCX

SCX - LESS ””””—— SCX C(s) - LESS LoRA adapter SCX - LESS SCX
state-value r(s)(s)L(s)[1-D(s)]

B.3 S2L (SmallToLarge)

- Yang et al., “SmallToLarge (S2L): Scalable Data Selection for Fine-tuning Large Language Models by Summarizing Training Trajectories of Small Models”, NeurIPS 2024
- <https://arxiv.org/abs/2403.07384>
- <https://github.com/BigML-CS-UCLA/S2L>

40-100

1. Pythia-70M 2. checkpoint loss → loss 3. K-means ~100 →

- 11% MathInstruct - 4.7%6 - MATH 50K Phi-2 16.6%

loss → S2L

SCX 1. vs S2L loss SCX + 2. vs S2L SCX 3. vs S2L SCX

SCX - S2L ——loss SCX - S2L SCX - S2L K-means K ~100SCX

B.4

B.4.1 InstructMining

- Cao et al., 2024
-

B.4.2 “Maybe Only 0.5% Data is Needed”

- Chen et al., 2023/2024
-

B.4.3 “Rethinking Data Selection at Scale: Random Selection is Almost All You Need”

- 2024
-

B.4.4 “A Critical Look at Targeted Instruction Selection” (Nayak et al., 2025)

- <https://arxiv.org/abs/2602.14696>
- LESS round-robin optimal transport

B.4.5

- “A Survey on Data Selection for LLM Instruction Tuning” (Wang et al., JAIR 2025)
- “Unleashing the Power of Data Tsunami” (Qin et al., TMLR 2024)
- <https://arxiv.org/abs/2408.02085>

IFD GPT k-center loss

SCX /SCX + +

C. Active Learning

C.1 DP-GEN (Deep Potential Generator)

- Zhang et al., Phys. Rev. Materials 3, 023804 (2019)
- <https://deepwiki.com/deepmodeling/dpgen/>
- <https://github.com/deepmodeling/dpgen>

Deep Potential DFT

4 DeePMD ensemble $\rightarrow$ Model Deviation

$$\hat{f}_{\max} = \max_i \langle ||f_i - \langle f_i \rangle||^2 \rangle$$

trust level_f [lower, upper] $\rightarrow$ _f < lower $\rightarrow$ _f > upper $\rightarrow$

SCX 1. **Ensemble** vs DP-GEN ensemble SCX + + 2. **vs** DP-GEN 0.12 eV/ASCX ””—— 3. **Exploration** vs DP-GEN ensemble explorationSCX - -

SCX - DP-GEN ” ensemble ”””SCX C(s) - DP-GEN 4 SCX - DP-GEN SCX

C.2 MTP / MLIP-3 — Extrapolation Grade + D-Optimality

- MLIP-2 / MLIP-3 (<https://gitlab.com/ashapeev/mlip-2>)
- Shapeev, “Moment Tensor Potentials”, 2016

Moment Tensor Potential (MTP) DFT

1. **Extrapolation Grade (cfg)** MTP(cfg) = $\max_j |c_j|c = (b \dots b_m)$
 A^1A active set > 1 2. **D-Optimality (MaxVol)**

MD $> _th 10 > 1 \rightarrow$ D-optimality $\rightarrow$ DFT $\rightarrow$

SCX 1. **vs** MTP extrapolation grade SCX 2. **vs** D-optimality SCX 3. **vs** MTP SCX

SCX - D-optimality extrapolation””SCX r(s) - D-optimality SCX N(s)L(s) - Clustering-enhanced local D-optimality, 2025 D-optimality —— SCX

C.3 DIRECT Sampling (DImensionality-Reduced Encoded Clusters with sTratified)

- “DImensionality-Reduced Encoded Clusters with sTratified (DIRECT) sampling”, Nature, 2024
- <https://www.nature.com/articles/s41524-024-01227-4>

MLIP

1. 2. 3. stratified sampling

SCX DIRECT $\rightarrow \rightarrow$ SCX ””

1. vs DIRECT /SCX r(s) 2. vs DIRECT SCX $\longrightarrow \rightarrow \rightarrow$ 3. vs DIRECT

SCX DIRECT ”” SCX SCX DIRECT $\longrightarrow$ DIRECT

C.4 DIRECT: Deep Active Learning under Imbalance and Label Noise

- Zhang et al., “DIRECT: Deep Active Learning under Imbalance and Label Noise”, 2024
- <https://arxiv.org/abs/2312.09196>

1D

SOTA >60%

SCX DIRECT softmax SCX SCX

C.5 QActor — Noise-Aware Active Sampling

- Ghiassi et al., “QActor: On-line Active Learning for Noisy Labeled Stream Data”, ACML 2021
- <https://arxiv.org/abs/2001.10399>

1. Quality Model 2. Active Querying oracle

30%-80% 6%

SCX - QActor "" vs - SCX NoiseScore(x_i) = r_i (1/(s_i)+) [1 - C(s_i)]""""
 - QActor SCX +

D.

D.1

/

Data Shapley	
TMC-Shapley	SCX
KNN-Shapley	()
Influence Func	
TRAK / LoGra	
LESS ()	
S2L (loss)	EGP()
DCLM ()	DIRECT()
InstructMining	DP-GEN(ensemble)
Random	MTP D-opt

- SCX """"EGP DIRECT MTP D-optimality DP-GEN - Data ShapleyIn-
 fluence FunctionsLESSS2L——/

D.2

Data Shapley	Y	Y	N	N	N	N
TMC-Shapley	Y	Y	N	N	N	N
KNN-Shapley	Y	Y	N	N	N	N
CHG Shapley	Y	Y	N	N	Partial	N
Influence Func	Y	Y	N	N	N	N
TRAK	Y	Y	N	N	N	N

LoGra/LogIX	Y	Y	N	N	N	N
LESS	N	Y	N	N	N	N
S2L	N	Y	N	N	N	N
DCLM	N	Y	N	N	N	N
DP-GEN	N	Y	Y(ensemble)	N	N	N
MTP D-opt	N	Y	N	N	N	N
DIRECT(Nature)	N	Y	N	Y()	N	N
QActor	N	Y	N	N	Y()	N
SCX	Y	Y	Y	Y	Y	Y

SCX

D.3

Data Shapley (MC)	$O(N^2 \times \text{train})$	N/A	(<10)
KNN-Shapley	$O(n \log n)$	$O(n)$	()
TRAK	$O(N \times d \times k)$	$O(dk^2)$	
LoGra	$O(N \times (dk))$	$O((dk))$	
LESS	$O(N \times d \times n_{\text{val}})$	$O(d \times n_{\text{train}})$	
S2L	$O(N \times T) + \text{K-means}$	$O(T \times n_{\text{small}})$	
DP-GEN	$O(4 \times N \times \text{fwd})$	$O(4 \times \text{train})$	(4 model ensemble)
MTP D-opt	$O(N \times d^2)$	$O(N \times d^2)$	
SCX	$O(N \times d + K \times N)$	$O(N \times d + \text{K-means})$	

SCX $O(N \times d)$ $O(N \times d + K\text{-means})$ $O(K \times M)$

E. SCX

E.1 SCX state-conditioned

SCX

1

Data ShapleyInfluence FunctionsLOO””——SCX Proposition 1 s s

2

MoE””SCX Proposition 1 3 A s B s B

MLIP ——””

3

max-residualuncertainty sampling/SCX Proposition 2 $H = H_{\text{learnable}}$
 H_{noise} “” “”

SCX - + + $\rightarrow$ - + + $\rightarrow$ /

4

Data ShapleyLESS/S2LLLM DP-GEN/D-optQActorSCX + +

E.2 SCX

1 Data ShapleyInfluence FunctionsLESSS2LTRAK /MLIP SCX

2 DCLMDIRECTD-optimality perplexitySCX $r(s)$ (s) $L(s)$ $[1 - D(s)]$ $r, L,$
 SCX_m

3 Data ShapleyLESSS2LDP-GEN /DP-GEN ensemble SCX MACE + NEP
 + ACE

4

- Data Shapley Influence Functions
- LESS S2L
- DP-GEN ensemble ””””
- MTP D-optimality

SCX ””””Proposition 2

5 SCX expert routing ——””””//” MoE

E.3

S2L - - S2L SCX - S2L $\rightarrow$ 4-7 SCX

LESS - ”” - LESS SCX - LESS

DP-GEN - - DP-GEN ensemble SCX + - DP-GEN

DIRECT (Nature 2024) - + - DIRECT SCX + - DIRECT SCX

E.4 SCX

1. /
2. SCX 2
3. R_m(s) L(s)
4. /
5. heuristic SCX +

E.5



7.

Data Valuation

1. Ghorbani & Zou, “Data Shapley: Equitable Valuation of Data for Machine Learning”, ICML 2019. <https://arxiv.org/abs/1904.02868>
2. Jia et al., “Towards Efficient Data Valuation Based on the Shapley Value”, AISTATS 2019. <https://arxiv.org/abs/1902.10275>
3. Li et al., “CHG Shapley: Efficient Data Valuation and Selection”, 2024. <https://arxiv.org/abs/2406.11730>

4. “Fast-DataShapley: Neural Modeling for Training Data Valuation”, 2025. <https://arxiv.org/abs/2506.05281>
5. “Asymmetric Data Shapley for Structure-Aware Valuation”, 2025. <https://arxiv.org/abs/2511.12863>
6. “Geometric Data Valuation via Leverage Scores”, 2025. <https://arxiv.org/abs/2511.02100>
7. “Faithful Group Shapley Value”, NeurIPS 2025. <https://neurips.cc/virtual/2025/sandiego/poster/115078>
8. Park et al., “TRAK: Attributing Model Behavior at Scale”, ICML 2023. <https://arxiv.org/abs/2212.02522>
9. Zheng et al., “D-TRAK: Intriguing Properties of Data Attribution on Diffusion Models”, ICLR 2024. <https://github.com/qwkkkkk/D-TRAK>
10. Choe et al., “What is Your Data Worth to GPT? LLM-Scale Data Valuation with Influence Functions”, NeurIPS 2025. <https://arxiv.org/abs/2405.13954>
11. “Generalized Group Data Attribution”, 2024. <https://arxiv.org/abs/2410.09940>
12. “Kairos: Scalable Model-Agnostic Data Valuation”, 2025. <https://arxiv.org/abs/2506.23799>

LLM Data Selection

13. “DataComp-LM: In search of the next generation of training sets for language models”, NeurIPS 2024. <https://arxiv.org/abs/2406.11794>
14. Xia et al., “LESS: Selecting Influential Data for Targeted Instruction Tuning”, ICML 2024. <https://arxiv.org/abs/2402.04333>
15. Yang et al., “SmallToLarge (S2L): Scalable Data Selection for Fine-tuning LLMs”, NeurIPS 2024. <https://arxiv.org/abs/2403.07384>
16. Qin et al., “Unleashing the Power of Data Tsunami: A Survey on Data Assessment and Selection for Instruction Tuning”, TMLR 2024. <https://arxiv.org/abs/2408.02085>
17. Nayak et al., “A Critical Look at Targeted Instruction Selection”, 2025. <https://arxiv.org/abs/2602.14696>
18. Wang et al., “A Survey on Data Selection for LLM Instruction Tuning”, JAIR 2025.

Active Learning

19. Zhang et al., “DP-GEN: Deep Potential Generator”, Phys. Rev. Materials 3, 023804 (2019). <https://github.com/deepmodeling/dpgen>
20. Shapeev, “Moment Tensor Potentials”, 2016. <https://gitlab.com/ashapeev/mlip-2>

21. “DIRECT: DIMensionality-Reduced Encoded Clusters with sTratified Sampling”, Nature 2024. <https://www.nature.com/articles/s41524-024-01227-4>
22. Zhang et al., “DIRECT: Deep Active Learning under Imbalance and Label Noise”, 2024. <https://arxiv.org/abs/2312.09196>
23. Ghiassi et al., “QActor: On-line Active Learning for Noisy Labeled Stream Data”, ACML 2021. <https://arxiv.org/abs/2001.10399>

Tools

24. pyDVL (Python Data Valuation Library). <https://github.com/aai-institute/pyDVL> — v0.10.0, 2024
25. LogIX / LoGra. <https://github.com/sangkeuncho/LogIX> — 2024